MLX91804 firmware example “bootstrap_parsing”

This firmware example implements the following endless sequence of bootstraps:

1) POR (after power on)
2) Wake-Ups from the Wake-Up timer (Warm boots) => 8 times with delay 500 ms, deep sleep between
3) the wake-ups
4) SW resets => 5 times
5) Resets from Absolute Watchdog timer (AWD) => 2 times
6) Go to item 2 above

The function **StringOut_via_UART** in the **main.c** is used to send strings via UART on the GPIO0 to inform about the bootstrap type.

The wake-up number is stored in the word **Warm_Boot_Counter** located in the scratchpad to retain its value between wake-ups during the deep sleep.

The resets' number is stored in the NOINIT_MEM RAM in the variable **Reset_Counter** that is assigned either in the section NOINIT_MEM or at the last word of the RAM for the case when the application is compiled and linked together with the LF bootloader. As a result, the variable **Reset_Counter** is not cleaned neither by HW (unlike of the scratchpads after resets), nor by the pre-main firmware (when it cleans the global C-variables after the cold boot) => it stores data between resets.

There is a picture explaining the example sequence (LF bootloader is not used):